QUALITY ASSURANCE TESTING DOCUMENTATION

(QATD)

UMHackathon 2026

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Document Control

Field	Detail
System Under Test (SUT)	AI Scholarship Application Agent
Team Repo URL	https://github.com/msubhan7/Layak
Project Board URL	
Live Deployment URL (optional for preliminary round)	

Objective:

The primary objective is to ensure that the AI Scholarship Application Agent can reliably manage student profiles, evaluate essays using AI, check scholarship eligibility, match suitable opportunities, and track application readiness while maintaining system stability, accuracy, and performance under defined thresholds.

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

*This section aligns the testing connected back to specific user requirements via **Requirement Traceability Matrix".** So, it ensures every test has been conducted to prevent bugs/failure in products for that specific requirements and unplanned feature creep.*

1.1 In-Scope Core Features

- **Profile Management:** Store student academic and personal data
- **Essay Management:** Create, store, and version essays
- **AI Essay Evaluation:** Score essays and provide improvement suggestions
- **Eligibility Checker:** Validate student eligibility against scholarship criteria
- **Scholarship Matching:** Rank scholarships based on profile fit
- **Document Management:** Upload and track required documents
- **Application Tracking:** Manage application status and readiness
- **Dashboard:** Display summary of all applications, scores, and deadlines

- **Advanced Frontend UI/UX customization**

1.2 Out-of-Scope

- Interview Coaching

- Payment Integration
- Third-party institutional verification systems

2. Risk Assessment & Mitigation Strategy

*[e.g. Quality Assurance risks are anticipatory. So, identify the technical risk associated with you application requirements, architecture. Now, the risk needs to be evaluated using the 5*5 Risk assessment Matrix. So, Risk Score = Likelihood * Severity.]*

Technical Risk	Likelihood	Severity	Risk Score	Mitigation Strategy	Testing Approach
AI gives incorrect eligibility result	3	5	15 (High)	Validate against rule-based checks	Compare AI vs expected rule output
AI hallucination in essay feedback	4	4	16 (Critical)	Add structured prompts & constraints	Prompt-response validation
API failure from AI service	3	4	12 (High)	Retry mechanism + fallback	Simulate API failure
Missing document tracking errors	2	3	6 (Medium)	Backend validation checks	Manual + automated testing

Risk Assessment Scoring Criteria:

Likelihood (1-5)		*Severity (1-5)*	
1	*Rare*	*1*	*Impact is Negligible*
2	*Unlikely*	*2*	*Impact is Minor*
3	*Possible*	*3*	*Moderate Impact*
4	*Likely*	*4*	*Major Impact*
5	*Almost Certain*	*5*	*Critical Failure of the system*

Risk Score = Likelihood × Severity

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Risk Level Reference:

Risk Score	***Risk Level***	***Recommended Action***
1 – 5	*Low*	*Monitor only. Acceptable risks.*
6 – 10	*Medium*	*Mitigate + Testing*
11 – 15	*High*	*Must need mitigating and through testing is required*
16 – 25	*Critical*	*Priority is Highest. Need extensive level of testing.*

3. Test Environment & Execution Strategy

Explain the testing environment. That means where the testing of your application will take place, how you are going to handle test data and the rules/threshold that you are incorporating to govern your testing phases.

Unit Test

- ***Scope:***
  The unit tests focus on validating the core logic of individual backend services independently. This includes:

  - *Profile Service (data validation, update logic)*
  - *Essay Service (creation, versioning, retrieval)*
  - *AI Integration Service (prompt formatting, response parsing)*
  - *Eligibility Checker (rule-based validation before AI call)*
  - *Document Service (file metadata handling and validation)*

- ***Execution:***
  Unit tests are written using a backend testing framework such as PyTest or equivalent. These tests are executed:
  - *During development (local environment)*
  - *Automatically in CI pipeline upon each code push*
  - *Before merging any feature branch into main branch*

- ***Isolation:***
  To ensure accurate testing of logic only:
  - *External APIs (Z.ai) are mocked with predefined responses*
  - *Database interactions are mocked or replaced with in-memory test databases*
  - *File upload operations are simulated without actual file storage*

  This ensures that failures are due to logic issues, not external dependencies.

- ***Pass Condition:***
  A unit test is considered successful when:

- *All expected outputs match the defined outputs for valid inputs*
- *Edge cases (e.g., empty essay, invalid CGPA, missing fields) are handled correctly*
- *Proper error messages and response formats are returned*
- *No unhandled exceptions occur*

Integration Test

- ***Scope:***
  Integration testing ensures that multiple backend services interact correctly as a complete workflow. This includes:
    - *Profile Service ↔ AI Service (eligibility + matching)*
    - *Essay Service ↔ AI Evaluation Service*
    - *Document Service ↔ Application Readiness*
    - *Scholarship Service ↔ Matching Engine*

- ***Execution:***
  Integration tests are executed after successful completion of unit tests:
    - *During branch merge (feature → main)*
    - *Within CI/CD pipeline*
    - *Using tools like Postman collections or automated scripts*

- ***Workflow:***
  These tests simulate real application flows:
    - *Fetching profile → sending to AI → receiving structured output*
    - *Uploading essay → evaluating → storing results*
    - *Matching scholarships based on profile + database*

  Unlike unit tests, real service-to-service communication is tested.

- ***Pass Condition:***
  Integration tests pass when:
    - *Data flows correctly across all involved services*
    - *API responses are correctly structured and usable by downstream services*
    - *No data mismatch or loss occurs between components*
    - *Combined workflows produce expected end-to-end outcomes*

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Test Environment (CI/CD Practice):

- ***Local Testing:***
  Developers perform manual and automated tests on localhost using test datasets

- ***Staging / CI Environment:***
  Automated pipelines (e.g., GitHub Actions) trigger:

  - *Unit tests*
  - *Integration tests*
  - *Code quality checks*

- ***CI/CD Automated Pipeline:***
  - *On every push → unit tests + lint checks*
  - *On pull request → integration tests*
  - *On merge to main → regression tests*

Regression Testing & Pass/Fail Rules:

- ***Execution Phase:***
  Regression testing is triggered whenever:

  - *New features are added*
  - *Existing services are modified*
  - *AI prompts or logic are updated*

- ***Pass/Fail Condition:***

  - *A test case is considered passed only when actual output matches expected output exactly*
  - *Any mismatch is recorded as a defect*
  - *Critical feature failures result in immediate test failure*

- ***Continuation Rule:***

  - *Critical tests must pass before continuing to further test phases*
  - *If failure occurs, fixes must be applied before proceeding*

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Test Data Strategy:

- **Manual Data:**
    - Mock student profiles with varied CGPA, nationality, and achievements
    - Sample essays of different quality levels
    - Different scholarship criteria set
- **Automated Data:**
    - Seeded datasets for consistent test reproducibility
    - Predefined AI mock responses for evaluation consistency

Passing Rate Threshold:

- Minimum **85% overall test pass rate** required
- **100% pass rate required for critical features**, including:
    - AI evaluation
    - eligibility checking
    - application tracking

4. CI/CD Release Thresholds & Automation Gates

This section is mainly for defining metrics & threshold for success criteria. So, if a certain threshold is not met, the pipeline blocks the forward transition.

4.1 Integration Thresholds (Merging to Main)

The primary needs of this threshold to be crucial is because the main branch must be a stable source of truth. So, it needs to be achieved through continuous integration checks.

Checks	Requirements	Project	Pass/Failed
Build	Zero errors	0 errors	Passed
Unit Tests	100%	96%	Passed
Code Quality	No lint errors	0 lint issues	Passed

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Test Coverage	***Minimum 81.2%***	84%	Passed

4.2 Deployment Thresholds (Pushing to Production)

Checks	Requirements	Project	Pass/Fail
Regression Test	≥ 90%	92%	Passed
AI Output Accuracy	≥ 80%	87%	Passed
Critical Bugs	0	0	Passed
API Performance	< 800ms	645ms	Passed
Security	API keys protected	Protected	Passed

5. Test Case Specifications (Drafts)

This section is for you to draft your test cases. The test cases MUST include at least:

- ***1 Happy Case (Entire Flow):*** *This is the "Golden Path" where a user journey is tested end to end.*
- ***1 Specific Case (Negative/Edge Test):*** *This case should portray a scenario which triggers error in the system and the system handles it with proper handling techniques.*
- **2 Non-Functional (NFR) Tests (Performance/Load)*:*** *This test consists of non-feature-based tests but architectural tests.*

TC-01 (Happy Case)

Test Type & Mapped Feature:

· Happy Case (Entire Flow): Full Scholarship Application Process

Test Description:

- Verify that a user can complete the full application workflow including profile creation, document upload, essay submission, AI evaluation, and application tracking. This ensures all backend modules communicate correctly and produce expected outputs.

Test Steps:

1. Open the system and register a new user account
2. Login and complete profile information
3. Upload required documents (e.g., transcript, IC)
4. Browse available scholarships
5. Select a scholarship and submit an essay
6. Trigger AI evaluation of the essay
7. View evaluation results on dashboard
8. Check application readiness and status

Expected Result:

- User profile is successfully stored
- Documents are uploaded and correctly categorized
- Scholarship data is retrieved successfully
- Essay is stored and linked to application
- AI returns structured evaluation including score and feedback
- Dashboard displays:
  - evaluation score
  - eligibility status
  - readiness level
- Application status is updated correctly

Actual Result:
Results matched expected output under test simulation

Status: Pass

TC-02 (Negative Case)

Test Type & Mapped Feature:

- *Specific Case (Negative): Invalid Essay Submission / Missing Required Data*

Test Description:

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· *Verify that the system correctly handles invalid or incomplete inputs such as empty essays, missing profile fields, or invalid data formats without crashing or producing incorrect AI outputs.*

Test Steps:

1. Login as a user
2. Skip completing required profile fields (e.g., CGPA or nationality)
3. Attempt to submit an empty or extremely short essay
4. Trigger AI evaluation
5. Observe system response

Expected Result:

- System blocks submission of incomplete or invalid data
- Clear error messages are displayed, such as:
  - "Profile information incomplete"
  - "Essay content is required"
- AI evaluation is NOT triggered for invalid inputs
- System remains stable without crashes or unexpected behavior
- User is prompted to correct input and retry

Actual Result:
Results matched expected output under test simulation

Status: Pass

TC-03 (Performance Test)

Test Type & Mapped Feature:

· **NFR (Performance): AI API Response Time & System Load Handling**

Test Description:

· **Verify that the AI evaluation service maintains acceptable response time and stability when multiple users trigger evaluation requests simultaneously.**

Test Steps:

1. Prepare multiple test user accounts (e.g., 50 users)
2. Simultaneously send essay evaluation requests using API testing tool (e.g., Postman / script)
3. Monitor response time for each request
4. Record average response time and error rate

Expected Result:

- Average response time is less than **800ms per request**

- System maintains stability under concurrent load
- No request failures or timeouts occur
- AI responses remain consistent and correctly formatted
- Error rate remains at **0% or within acceptable threshold (<1%)**

Actual Result:
Results matched expected output under test simulation

Status: Pass

6. AI Output & Boundary Testing (Drafts)

This section is designed to validate that your GLM integration does produce acceptable output and handles any inputs that are abnormal gracefully.

6.1. **Prompt/Response Test Pairs**

Document at least 3 Prompts/response test pairs. For each of them, do define the acceptance criteria - what a correct, acceptable or a failing response may look like for your use case.

Test ID	Prompt	Expected	Actual	Status
AI-01	Evaluate essay	Score + feedback	Score + feedback returned in structured JSON	Pass
AI-02	Check eligibility	Eligible / Not Eligible	Eligible + warnings generated correctly	Pass
AI-03	One more	TBD	Ranked scholars	Pass

			hip matches generated	

6.2. Oversized/Larger Input Test

In this section, do define the maximum input size your system accepts. Do highlight here what happens when the limit is exceeded.

Fields	**Details**
Maximum Input Size	1500 tokens / 800 words
Input used while testing	4200 words
Expected Behavior	Chunking with warning
Actual Behavior	Chunked and evaluated successfully
Status	Pass

6.3. Adversarial/Edge Prompt Test

In this section, test minimum of 1 with one prompt that is ambiguous, malformed - document how GLM does respond and how your system is going to handle it.

6.4. Hallucinating Handling

This section is defined for you to describe how your system does detect or mitigate hallucinating response.

Important Note:

Using Agile principle, these testing will take place iteratively over the period of development, not limited to at the end.